TVAC

Tokenization Value-Add Calculator

INDEPENDENT. STRUCTURED. FOR REAL-WORLD DECISIONS.

Deep Assessment Report

Tokenized Commercial Paper Issuance & Lifecycle — United States

A case-specific assessment report for tokenization viability and value creation.

November 2, 2026

A TVAC assessment by Astoval

CLARITY
BEFORE
COMMITMENT.

About this report

TVAC evaluates whether tokenizing a specific asset / structure creates net added value, using the five-factor model:

> **Added Value = (New Opportunities (NO) + Cost Savings (CS) + Risk Reduction (RR)) – (Tokenization Costs (TC) + New Risks (NR))**

Where this report fits

TVAC is designed for the very early decision stage of a tokenization initiative. It tests whether the proposed use of tokenization appears likely to create genuine added value before significant resources are committed to legal structuring, issuance-platform or provider selection, detailed technical architecture, or implementation. A positive verdict is therefore not implementation approval; it indicates that the case may justify deeper legal, commercial, technical and operational work.

Case description summary

The initiating organization is a US-based corporate issuer with an existing, recurring commercial paper (CP) program used for short-term working capital. The proposed design seeks to introduce a tokenized CP note format (economically equivalent to the organization's existing CP) for professional and accredited investors only. Today the program relies on multiple intermediaries, manual confirmations, repeated reconciliations between the issuer’s treasury team, issuing/placing agents, custodians, and investors, and a settlement cycle that creates operational friction and occasional settlement fails. The scope is deliberately conservative: a single US issuance + distribution perimeter, whitelisted participation, and institutional-grade custody. This is not a retail product and not a public exchange initiative.

The full case description and structured input record used for this assessment are reproduced in Appendix E.

Generated	November 2, 2026
Methodology	TVAC — Tokenization Value-Add Calculator
Published by	Astoval ApS

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01 Executive summary

TOKENIZATION VALUE VERDICT

Viable

ADDED VALUE

+9.00

plus 16.00 **minus** 7.00

Purpose and tokenization test

Project objective: Explicit objective: improve post-trade efficiency and reduce operational and settlement friction in an existing recurring US commercial paper funding program by introducing a tokenized CP note format that remains economically equivalent to the current CP instrument.

Intended beneficiaries: The primary intended beneficiary is the US corporate issuer and its treasury function through lower reconciliation effort, fewer manual exceptions, and tighter control over issuance and maturity processing. Professional and accredited investors may benefit from cleaner servicing and custody workflows, but broad access expansion is not an objective. Unknowns remain around how benefits are split contractually across agents, custodians, and other infrastructure providers.

Conventional comparator: The issuer's existing recurring non-tokenized US commercial paper program, including current treasury coordination, agents, custodians, investor servicing, manual confirmations, repeated reconciliations, and the present settlement process.

What tokenization must improve: Does the token layer improve the same US commercial paper funding workflow enough to reduce recurring reconciliations, manual confirmations, settlement exceptions, and lifecycle-processing friction beyond what the existing CP process or a non-token workflow upgrade would deliver?

ASSESSMENT RATIONALE

The tokenization case scores 9.00 (Viable); this verdict assesses incremental value rather than the underlying asset's investment merits. The token-specific mechanism is operational rather than distribution-led: a proposed shared operating record, automated issuance and lifecycle events, controlled whitelisted transfers, and DvP-style settlement controls for a US-only institutional perimeter. The main benefit should accrue to the issuer if duplicated legacy steps are truly retired. Incremental benefit is plausible, but the exact degree still depends on record authority, real process retirement, provider economics, and operating evidence.

Important interpretation note: This verdict concerns tokenization added value only. A Viable or Conditional Go verdict does not mean that the underlying asset is a good investment, that projected returns are realistic, that credit risk is acceptable, or that the instrument is suitable for any investor.

Yield neutrality note: The stated coupon, yield, interest rate, or expected return has not been treated as tokenization value in itself. TVAC does not assess whether projected returns are realistic, sustainable, appropriately priced for risk, or suitable for investors. Yield may be relevant to investor demand, but it does not by itself demonstrate that tokenization adds value.

KEY TAKEAWAYS

01 **Key findings:** Incremental savings depend on whether the tokenized tranche truly retires manual reconciliations, confirmations, and exceptions on a recurrent basis and whether those savings exceed token-layer operating costs.

02 **Key constraints:** Relevant burden is the additional whole-life resource cost of the tokenized CP workflow, including setup, custody, recordkeeping, governance, and provider dependence, and who ultimately bears those costs.

03 **Decision posture:** Run a like-for-like operating comparison between one tokenized CP issuance workflow and the current non-tokenized CP process, covering confirmations, reconciliations, exception handling, settlement repair, and maturity servicing; continue if the tokenized model demonstrably retires recurring steps and clarifies control responsibility, redesign if it merely adds a second operating layer, and stop if provider economics or control uncertainty eliminate issuer-side advantage.

04 **Impact:** For the initiating stakeholder, the positive Added Value of 9.00 supports further evaluation, conditional on the documented evidence and unresolved costs and risks. Incremental opportunity comes mainly from issuance-lifecycle optimization within a recurring US CP workflow, not from broader distribution or secondary trading.

02 Asset class & market context

This is primarily an issuance-and-lifecycle optimisation case within an established recurring debt-issuance programme. Tokenization value depends on measurable improvements in allocation, settlement, authoritative position records, reporting, interest or discount administration, maturity processing and exception handling — not on new capital access or assumed secondary liquidity.

Compare the proposed token-based workflow with an already digital non-tokenized process that serves the same participants, obligations and transaction events. Electronic onboarding, ordinary databases and automated approvals are available in that comparator and do not by themselves establish incremental tokenization value. A shared token state may improve handoffs, restrictions or auditability when it replaces existing work rather than duplicating it.

Those gains must be weighed against integration, authoritative-record consistency, failed-event recovery, cash-leg dependencies where applicable, and ongoing provider and governance obligations. The critical evidence is the difference in processing effort, exceptions, controls and economic benefit for comparable events before and after adding the token layer. No local market volume, uptake, cost level or regulatory status is inferred without verified sources.

U.S. tokenized-securities infrastructure has moved beyond purely conceptual treatment. In January 2026, SEC staff described issuer-sponsored tokenized securities as a recognized implementation model and stated that securities capable of tokenization include notes. In 2026, DTCC also moved its DTC tokenization service into limited production activity for selected DTC-custodied assets.

These developments show institutional infrastructure progress, but they do not establish a mature or liquid U.S. market specifically for tokenized commercial paper.

What this context means for the case

Structural factors supporting the case:

- The existing recurring issuance programme provides a concrete conventional baseline against which operational improvement can be tested.
- The stated professional-investor and controlled-transfer perimeter keeps the value case focused on workflow improvement rather than speculative liquidity expansion.
- SEC staff explicitly describes issuer-sponsored tokenized securities as a model in which DLT can be integrated into the issuer or agent ownership-record system, and notes are among the security types that can be tokenized.
- DTCC reported 2026 production activity using DTC-tokenized assets, providing evidence that U.S. market infrastructure for tokenized securities is progressing beyond paper design and isolated experimentation.

Structural factors that still constrain the case:

- The authoritative record, custody or registry responsibilities, cash-leg finality and recovery procedures must be validated before settlement-sensitive benefits can be credited.
- The value case weakens materially if the tokenized tranche remains a permanent parallel process instead of retiring enough legacy issuance and servicing steps.
- Any claimed settlement or automation benefit remains dependent on a workable cash leg, finality, exception handling and reconciliation across the selected infrastructure.
- The verified sources do not quantify issuance volume, investor demand, liquidity, cost savings or adoption specifically for tokenized U.S. commercial paper.
- DTCC’s current tokenization milestones concern selected DTC-custodied asset classes and broader securities infrastructure; they should not be presented as evidence that commercial paper itself has reached comparable production adoption.

03 Analysis result

ADDED VALUE

+9.00

PLUS	MINUS
16.00	7.00

Purpose and tokenization test

Potential mechanism and evidence: The token-specific mechanism is operational rather than distribution-led: a proposed shared operating record, automated issuance and lifecycle events, controlled whitelisted transfers, and DvP-style settlement controls for a US-only institutional perimeter. The main benefit should accrue to the issuer if duplicated legacy steps are truly retired. Incremental benefit is plausible, but the exact degree still depends on record authority, real process retirement, provider economics, and operating evidence.

Decision-critical unknowns

- Which record is legally authoritative for ownership and lifecycle rights, and would any unresolved hierarchy force participants back into manual reconciliation?
- Will the tokenized CP tranche actually retire duplicated legacy controls, or will participants preserve parallel confirmations and records that dilute Cost Savings?
- How does the DvP-style workflow function against the stated T+1/2 payment process in both normal and failed-settlement scenarios?
- Are provider scope, pricing, and portability favorable enough that recurring issuer-side savings are retained rather than absorbed by the new operating stack?
- Can qualified US counsel confirm a realistic regulatory path for the final tokenized CP operating model and transfer-control design?

Verdict rationale:
This case is assessed as Viable because the current user-defined structure produces Added Value +9 (plus 16 minus 7), which places the case in the Viable band under the TVAC methodology. In practical terms, the strongest supportive driver is Cost Savings (6/10), while the main friction is Tokenization Costs (5/10). The cost case is stronger than in many tokenization projects because the issuer expressly intends to retire duplicated legacy steps for the tokenized CP tranche rather than maintain permanent parallel operations. This design still carries meaningful tokenization cost categories: workflow infrastructure, custody and recordkeeping support, legal and operational setup, and ongoing governance for the tokenized CP process. This rationale concerns tokenization added value only and does not validate legal, regulatory, tax, financial, investment, credit, yield, issuer, or implementation suitability.

Factor score profile

Factor	Score	Case-specific driver
New Opportunities	**5/10**	Incremental opportunity comes mainly from issuance-lifecycle optimization within a recurring US CP workflow, not from broader distribution or secondary trading.
Cost Savings	**6/10**	Incremental savings depend on whether the tokenized tranche truly retires manual reconciliations, confirmations, and exceptions on a recurrent basis and whether those savings exceed token-layer operating costs.
Risk Reduction	**5/10**	Incremental risk reduction must come from additional control quality versus the same CP comparator, especially fewer breaks, better auditability, and tighter settlement handling, not from comparing the design with a weaker crypto-native alternative.
Tokenization Costs	**5/10**	Relevant burden is the additional whole-life resource cost of the tokenized CP workflow, including setup, custody, recordkeeping, governance, and provider dependence, and who ultimately bears those costs.
New Risks	**2/10**	Relevant new risks are token-specific additions versus the current CP process, especially record-authority ambiguity, provider dependency, and operational-control mismatch between token and cash legs.

Added Value: 9.00 (plus 16.00 / minus 7.00)

What still needs to be verified:

These are the principal evidence gaps requiring independent verification before relying on this assessment for further decisions.

- No verified case-specific regulatory claim is activated; approved baseline context supports structured scoping, while case-specific legal application remains unresolved.
- Obtain written legal and control analysis of the token layer, authoritative instrument and investor records, participant and intermediary roles, custody or registry responsibilities, settlement finality and the operating perimeter.
- Document and validate the relationship between the tokenized workflow, authoritative issuer, agent, registry or custody records, security- and cash-leg finality, exception handling, rollback and recovery.

Regulatory status: United States · Case-specific applicability unverified

TVAC has identified relevant regulatory context for the stated issuance jurisdiction and case-specific legal questions, but it has not verified how the law applies to the final structure. For an existing or retrospective case, historical authorizations and ongoing obligations require independent validation; for a planned case, applicable requirements should be assessed before implementation.

Sources used and outstanding validation questions are summarized in Appendix A.5.

> The score becomes useful only when the structure behind it is coherent. The sections below test the value pathways, participant or investor perimeter, operating burden, regulatory path, custody and transfer fit, and—where material—the settlement and cash-leg design that must support the claimed benefits.

3.1 Value pathways

This section isolates the mechanisms through which tokenization could create incremental value over a realistic conventional alternative. It is not an assessment of whether the underlying asset or financing need is attractive in its own right.

- The core value case is not broader distribution or retail access; it is issuance-lifecycle optimization for an existing US commercial paper program already used for short-term working capital.
- The proposed tokenized CP note format could create incremental value by collapsing repeated post-trade handoffs between the issuer treasury team, agents,

custodians, and investors into a more unified operating record and message flow.

- The issuer states that the tokenized tranche would retire duplicated legacy steps rather than preserve permanent double-booking. That design choice is economically important because it turns automation from cosmetic digitization into a potential recurrent operating change.
- Because secondary liquidity is explicitly out of scope and transfers are tightly controlled, the value thesis depends mainly on better issuance, servicing, exception management, and settlement discipline within the United States perimeter.
- The primary beneficiary is the issuer rather than a trading venue or retail investor base, which aligns the case more closely with asset-owner economics than with a liquidity-marketing narrative.

3.2 Investor segmentation

Investor scope affects both upside and friction. A narrower professional or institutional audience can reduce distribution and compliance complexity, but it can also limit the access or liquidity benefit that tokenization is expected to create.

- The investor perimeter is intentionally institutional: professional and accredited investors only, with retail excluded and whitelisting plus issuer or agent-approved transfers.
- That segmentation supports control and reduces go-to-market complexity, but it also means New Opportunities should be understood as operational and servicing improvements rather than democratized access or liquidity creation.
- For US commercial paper, the relevant adoption question is whether eligible investors and their custodial workflows can process the tokenized note format with less friction and no unacceptable control gap versus the existing CP process.
- Because distribution remains within the United States and secondary liquidity is not a core objective, investor segmentation strengthens execution discipline even while it limits upside from broader placement reach.

3.3 Operational costs

This section distinguishes plausible process savings from the additional cost of introducing the token layer. A saving becomes a strong value driver only when it can be compared with the current process and is not offset by data, integration, provider or compliance work.

- The case can support recurring savings only if manual reconciliation, confirmations, and exception handling are actually retired for the tokenized tranche. The stated commitment to avoid permanent double-booking is therefore central to cost performance.
- Likely cost categories still include token workflow infrastructure, custody and recordkeeping support, legal and operational setup, and ongoing governance for the tokenized process. Their amounts are not stated.
- The recurring CP context is favorable for amortization because the same process can potentially be reused across repeated short-term issuances, but the extent of that benefit remains to be demonstrated against a conventional baseline.
- Institutional-grade custody and controlled transfers may increase some provider costs while still reducing whole-workflow effort if they replace duplicated downstream exceptions and reconciliations rather than sit beside them.

3.4 Regulatory path

This is regulatory scoping, not a legal opinion. The purpose is to identify the verified regulatory context and the design questions that materially affect the case, while making clear what still requires qualified jurisdiction-specific validation.

- TVAC has approved baseline regulatory context for United States, but it has not activated a verified case-specific local claim. The frameworks below are used for structured scoping only; applicability and current detailed conditions remain to be independently confirmed.
- At scoping level, a token representing debt, equity, fund interests or another investment exposure should be analysed under the U.S. federal securities-law framework. Tokenization does not itself determine the instrument's legal classification.
- Broker-dealer, alternative trading system, transfer-agent, custody and recordkeeping questions may arise according to the functions actually performed, the instrument classification and the proposed secondary-transfer model. A technology label does not decide those roles.

3.5 Custody & venue fit

Custody and venue design should be proportionate to the value thesis. If the case does not depend on continuous secondary trading, extra custody, venue or transfer complexity should be justified by a concrete operational, control or distribution benefit.

- The described custody model fits the case logic better than a self-custody model would, because the target holders are institutional and the value proposition centers on controlled operating workflows rather than crypto-native autonomy.
- OTC handling is also consistent with the stated objectives. Since liquidity is not the primary objective and secondary transfers are tightly controlled, a public trading venue is not necessary to support the present value case.
- The main fit question is therefore not market access breadth but whether the regulated custodian, agents, and recordkeeping workflow can support a recurring tokenized CP issuance process with fewer exceptions than the current setup.
- If the transfer model expands over time, broker-dealer, ATS, transfer-agent, custody, and recordkeeping questions may become more relevant. In the current design, venue complexity appears deliberately contained rather than central to the thesis.

3.6 Settlement & Cash-Leg Readiness

Settlement is assessed against the case's actual value logic, not against an assumption that tokenization requires digital money or T+0. The key question is whether the cash leg can operate reliably for the intended investors and operating model without unnecessary complexity.

- Why it matters: The claimed value depends partly on reliable DvP-style settlement and reduced settlement failures. The security leg and the corresponding cash movement therefore need to operate together reliably.
- Current position: the proposed payment and settlement approach is T+1/T+2 securities settlement. A settlement approach has been proposed, but its operational readiness has not been independently established. This does not establish legal or operational settlement finality.

- What must be proven: asset- and cash-leg finality, synchronization, failed-settlement handling, exception management and recovery. Until those points are evidenced, settlement-dependent benefits should remain conditional rather than treated as realized outcomes.

04 Stakeholder value & alignment

This chapter asks whether the parties needed to make the structure work are economically and operationally aligned. It focuses on the value each party receives, the burden and responsibility it carries, and the dependencies that could weaken the project's overall Added Value.

4.1 Value & alignment overview

The stakeholder map identifies the parties involved. Their distinct benefits, costs, responsibilities and alignment conditions are analysed once in the sections that follow.

US corporate issuer / treasury function
Initiator of the tokenized commercial paper workflow change and primary intended beneficiary

Professional and accredited investors in the United States
Eligible holders within a whitelisted, tightly controlled institutional perimeter

Issuing or placing agents, regulated custodian, recordkeeping and token-workflow providers
Delivery counterparties that enable issuance, custody, servicing, transfer controls, and maturity processing

4.2 US corporate issuer / treasury function

Value received

- The issuer could benefit from fewer recurring reconciliations, fewer manual exceptions, more standardized servicing, and better control over recurring CP issuance and maturity operations.

Costs, risks & responsibilities

- The issuer would bear much of the setup, governance, provider-management, and implementation effort, and it remains exposed if duplicated legacy steps are not actually retired.

Alignment & actions

- Alignment appears relatively strong because the case is expressly designed around issuer-side operational savings and control improvement rather than around speculative liquidity or retail distribution.

4.3 Professional and accredited investors in the United States

Value received

- Investors may receive cleaner servicing, more consistent position visibility within the proposed model, and potentially fewer settlement or processing issues than in the current CP workflow.

Costs, risks & responsibilities

- Their burden may include onboarding to the tokenized custody and transfer-control process and adapting internal handling for the tokenized CP note format. The scale of that burden is not stated.

Alignment & actions

- Alignment is functional rather than promotional: investor participation supports the issuer's operating redesign, but the case does not depend on investors receiving a new liquidity product or broader market access.

4.4 Issuing or placing agents, regulated custodian, recordkeeping and token-workflow providers

Value received

- These parties may gain new service revenue and a more standardized operating model if the tokenized CP format is adopted across recurring issuances.

Costs, risks & responsibilities

- They may need to integrate new workflows, support operating tests, and accept clearer responsibility boundaries. Their final commitments, service levels, and pricing are not stated.

Alignment & actions

- Alignment depends on whether provider terms help the issuer capture the operational gains. If providers preserve parallel processes or price the token layer inefficiently, alignment would weaken materially.

4.5 Who bears the costs and risks?

For this qualified high-volume institutional workflow, the initiating institution may carry most front-loaded integration and governance costs. That does not by itself mean the economics favour service providers. The relevant test is whether the initiating institution captures measurable cumulative savings and risk reduction once the workflow reaches run-state.

- A neutral or low near-term initiating-institution position should trigger an evidence and cost-allocation review, not an automatic rejection. It becomes a warning sign where the expected operational benefits are not demonstrated, provider fees absorb the realised gains, or key service obligations remain misaligned with value capture.
- Eligible counterparties should still receive clear evidence of operational controls, custody synchronisation, settlement finality and the allocation of exception-handling costs.
- Where operational benefits are evidenced and all stakeholders are positive, focus on disciplined scale-up, service continuity and ongoing measurement of realised value.

05 Factor breakdown – detailed explanation

This chapter explains the causal basis of the five scores: which facts, assumptions and evidence gaps create upside, and which ones introduce additional friction. The three positive factors ask what tokenization adds relative to a realistic conventional alternative; the two negative factors ask what extra cost or risk the token layer introduces. A low positive score therefore means limited or weakly evidenced tokenization benefit, while a low negative score means limited additional friction. The scores are calibrated methodology judgments, not measured outcomes or legal conclusions.

> **Added Value = (New Opportunities + Cost Savings + Risk Reduction) – (Tokenization Costs + New Risks)**

The factor profile shows a positive tokenization value case. The strongest upside comes from cost savings, while the principal friction is tokenization costs; the explanations below set out the case-specific mechanisms and evidence gaps behind those judgments. With an Added Value score of 9, the value case is clearly positive rather than marginal.

New opportunities (NO)

- The strongest incremental opportunity is not new investor reach but a more efficient issuance-lifecycle model for recurring US commercial paper, which is harder to replicate if the tokenized tranche genuinely shares a unified operating record across participants.
- Lifecycle automation for confirmations, accrual or coupon reporting, and maturity notices adds a second, smaller opportunity by turning routine CP servicing into a more standardized recurring process.
- Upside is intentionally capped because the design excludes retail distribution, public exchange trading, and a broad secondary-liquidity proposition.
- The opportunity case would strengthen only if the issuer can show that the tokenized workflow improves the recurring CP process beyond what a non-token workflow upgrade could achieve.

Cost savings (CS)

- The cost case is stronger than in many tokenization projects because the issuer expressly intends to retire duplicated legacy steps for the tokenized CP tranche rather than maintain permanent parallel operations.
- Savings are tied to identified cost drivers in the current US CP workflow: repeated reconciliations, manual confirmations, agent and custody operations per issuance, and manual exception handling.
- Because the program is recurring, any valid process simplification can potentially be reused across multiple issuances instead of being absorbed by a single one-off transaction.
- The remaining uncertainty is the absence of a documented conventional baseline and provider pricing comparison, so the savings thesis is well-formed but not yet demonstrated quantitatively.

Risk reduction (RR)

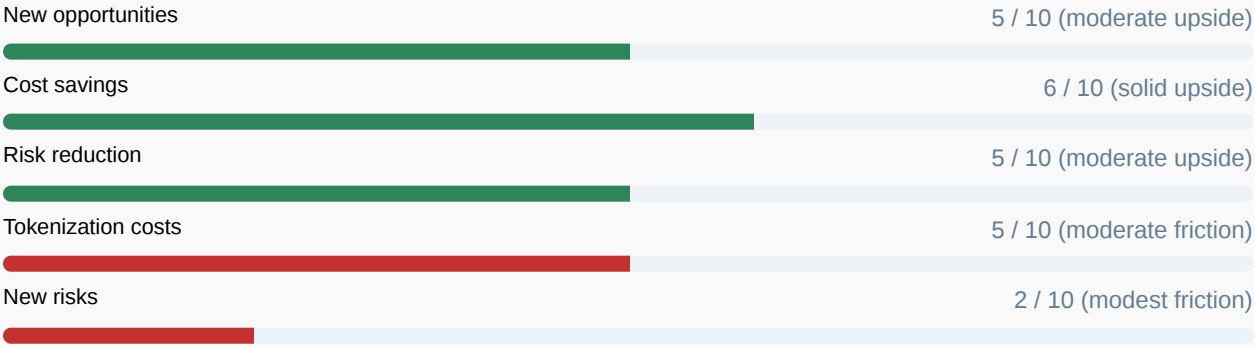
- The proposed tokenized commercial paper workflow could reduce operational risk by lowering reconciliation breaks and giving participants a cleaner shared position view in the proposed design.
- DvP-style controls and automated lifecycle messaging could also reduce settlement-fail and process-error exposure even if the cash leg remains on conventional T+1/2 rails.
- The narrow US-only, whitelisted, institutional perimeter helps contain distribution and transfer complexity relative to a broader cross-border or retail design.
- Risk reduction is still bounded by unresolved record-authority and exception-management questions, so the case supports meaningful but not absolute control improvement.

Tokenization costs (TC)

- This design still carries meaningful tokenization cost categories: workflow infrastructure, custody and recordkeeping support, legal and operational setup, and ongoing governance for the tokenized CP process.
- Recurring issuance may help spread those costs, but the extent of that benefit is not yet evidenced through quoted pricing or measured baseline comparison.
- The conservative institutional design reduces unnecessary complexity, yet it does not remove the need for careful integration between treasury, agents, custodians, and the tokenized note process.
- Costs remain manageable rather than excessive because the project avoids retail distribution, public market venue build-out, and a broad cross-border rollout in its initial scope.

New risks (NR)

- New tokenization-specific risks appear relatively limited because the issuer is not relying on open trading, retail usage, or a complex multi-jurisdiction distribution model.
- The main new exposures are provider dependency, record-authority ambiguity, and the possibility that DvP-style controls do not deliver the expected operational benefit against conventional cash settlement.
- These risks matter, but they are narrower than in many tokenized issuance cases because the transfer model is tightly restricted and the instrument remains economically equivalent to the existing CP program.
- Residual uncertainty remains around the final regulatory-role mapping and the exact control hierarchy across records and intermediaries, which is why low new risk should not be read as zero implementation risk.



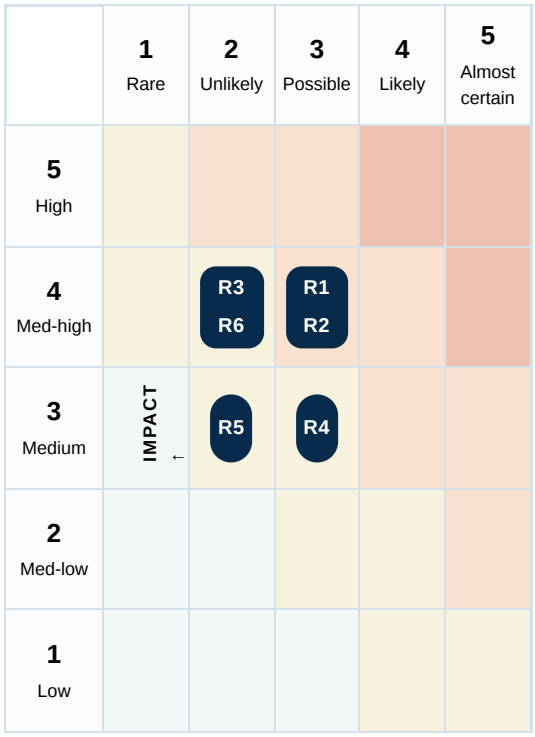
Plus (NO + CS + RR): 16

Minus (TC + NR): 7

Added Value (Plus – Minus): 9

06 Key risks & mitigation

This matrix prioritizes the report's identified risks by impact and likelihood. It is not a measured probability model; each position should be read together with the case-specific description and mitigation.



Risk zones are indicative prioritization bands, not measured probabilities.

PRIORITY RISKS

R1 **Misaligned incentives / vendor-capture risk**

Impact 4 · Likelihood 3

Mitigation: Use stage-gating based on legal/control validation and measurable operational-readiness evidence; link provider fees and scale-up to issuance completion, process retirement, service levels, reconciliation performance and lifecycle-delivery outcomes; preserve portability and exit rights if the provider arrangement changes.

R2 **Parallel process persistence**

Impact 4 · Likelihood 3

Mitigation: Document and enforce retirement of duplicated confirmations, reconciliations, and fallback steps for the tokenized CP tranche before repeated issuance use.

R3 **Record-authority ambiguity**

Impact 4 · Likelihood 2

Mitigation: Obtain written mapping of token records, custodian records, conventional documents, and exception-handling hierarchy for ownership and lifecycle events.

R4 **Partial DvP control benefit**

Impact 3 · Likelihood 3

Mitigation: Test release conditions, exception routing, and failure handling against the actual T+1/2 payment process before treating settlement-risk improvements as operationally achieved.

R5 **Provider dependency in recurring issuance**

Impact 3 · Likelihood 2

Mitigation: Secure clear service scope, operating responsibilities, pricing, and portability terms across issuance, custody, reporting, and maturity handling.

R6 **US regulatory-role uncertainty**

Impact 4 · Likelihood 2

Mitigation: Obtain case-specific US counsel analysis of instrument classification, transfer model, and intermediary-role implications for the final operating design.

07 Recommendations & next steps

7.1 Design priorities and decision gates

The decisive trade-off in this US commercial paper design is favorable but operationally conditional: the issuer appears positioned to gain real workflow value only if the tokenized tranche replaces enough of today's reconciliations, confirmations, and exception handling to justify the added infrastructure and governance layer. Overall verdict: Viable. This is sufficiently defined for a meaningful case-specific evaluation, but the lifecycle status is unspecified and several material design choices remain open, especially around record authority, DvP-style operating controls, and provider economics. The participant structure is narrow by design — a US issuer, professional and accredited investors, institutional-grade custody, OTC handling, and tightly controlled transfers — which helps contain Tokenization Costs and New Risks. Because trading liquidity is intentionally not the point, the case should be judged on whether the tokenized CP format improves recurring issuance operations for the issuer rather than on whether it creates a new market.

1. Make process retirement enforceable across the recurring CP workflow

Constraint in this case

The strongest value driver is the issuer's stated commitment to decommission duplicated legacy steps for the tokenized tranche. If treasury, issuing or placing agents, custodians, and investors continue to maintain parallel confirmations, reconciliations, or fallback records as a permanent operating norm, much of the expected gain becomes administrative layering rather than process improvement.

Required actions

- Define the exact legacy confirmations, reconciliations, and exception-handling steps that will be retired for each tokenized CP issuance.
- Assign one operating owner for each remaining control point across treasury, agents, custodian, and recordkeeping functions.
- Require each service provider in the tokenized tranche to support the target-state process rather than preserve a parallel legacy workflow by default.
- Build a fallback plan for operational incidents that is limited to exception use, not daily double-processing.

Expected effect

If permanent double-handling is actually removed, the case should improve mainly through stronger Cost Savings and clearer Risk Reduction, while preventing Tokenization Costs from being inflated by a redundant parallel stack. The confirming evidence is not narrative intent; it is an implemented operating model showing that the tokenized CP tranche uses fewer recurring manual steps than the current program.

2. Define record authority and control hierarchy before scaling recurring issuance

Constraint in this case

The proposed shared position record is central to the value thesis, but the current description does not state which record is legally authoritative for ownership and lifecycle rights or how conflicts between token records, custodian records, and conventional documents are resolved. Without that hierarchy, reconciliation savings may be partly reversed by control uncertainty.

Required actions

- Produce a written record map showing the role of the token record, custodian record, investor statements, and conventional CP documentation for the tokenized tranche.
- Document how issuance, transfer approval, maturity processing, and exception corrections are authorized and which party can direct each change.
- Align lifecycle notices and payment instructions to the same control hierarchy so that operational messaging does not outrun legal and custodial authority.
- Test edge cases such as failed settlement, holder correction, and maturity-day exceptions against the proposed record hierarchy.

Expected effect

This primarily protects Risk Reduction and keeps New Risks low by preventing a control mismatch between the token workflow and the conventional legal-operational backbone. The confirming evidence is a record-authority model that participants can actually operate without reintroducing manual reconciliations on disputed events.

3. Make the DvP-style design proportionate to the actual T+1/2 cash process

Constraint in this case

The case does not rely on instant settlement or public-market liquidity, which is helpful. Even so, the claimed settlement-control benefit depends on what "DvP-style" means in practice when the cash leg remains on conventional T+1/2 rails. If the securities leg is more controlled but the payment leg still depends on loosely coordinated manual release and repair processes, the benefit may be narrower than expected.

Required actions

- Specify the release logic for the tokenized CP note against the existing payment process, including who can authorize release, pause, or unwind steps.
- Document the exception path for late payment, mismatched instructions, and settlement repair on the actual cash rails to be used.
- Distinguish clearly between improved settlement control and true simultaneous finality so the design is evaluated against what it really delivers.
- Run end-to-end operating tests that include normal settlement and exception scenarios rather than only straight-through cases.

Expected effect

A proportionate DvP-style model can preserve Risk Reduction and some New Opportunities in process quality without requiring a more ambitious digital-cash architecture than this case needs. The confirming evidence is an operating design that reduces settlement breaks and repair effort on the issuer's real US payment process, not a generic claim of DvP readiness.

4. Lock provider economics and portability around the issuer's recurring-use case

Constraint in this case

The issuer is meant to be the primary beneficiary of the tokenized CP design. That requires the economic gains from recurring automation and cleaner post-trade control to accrue mainly to the issuer rather than being absorbed by provider fees, inflexible integrations, or vendor dependence that persists across every issuance cycle.

Required actions

- Obtain binding scope and pricing for issuance support, custody, recordkeeping, lifecycle messaging, and exception handling under the tokenized tranche model.
- Compare those provider costs directly with the current CP operating model, using the same workflow boundary and the same recurrent issuance assumptions.
- Negotiate data access, service continuity, and migration terms so the issuer is not locked into a costly operating stack if the process evolves.
- Tie provider responsibilities to measurable workflow outcomes such as reduced reconciliation handling and clearer maturity servicing, rather than paying only for technology presence.

Expected effect

This section mainly protects Cost Savings and prevents Tokenization Costs from being understated by incomplete commercial scoping. It also supports Asset Owner value capture by making sure the issuer keeps the operational benefit instead of funding a new layer whose economics mainly reward third parties. The confirming evidence is a like-for-like costed model showing that the recurring tokenized CP workflow is operationally superior on a whole-process basis, not just technically more modern.

Expected Outcome After Restructuring

The current case already supports a viable verdict because the tokenized commercial paper format is narrow, recurrent, and explicitly tied to a real post-trade pain point rather than to speculative liquidity claims. Even so, the result is strongest if the issuer treats this as an operating-model redesign rather than as an additional digital wrapper around the existing CP program.

Under revised assumptions, these changes could shift the Added Value profile and the verdict outcome — but this depends on execution, jurisdiction-specific constraints, evidence of the stated participant or user outcomes, and provider commitments. In practice, the key test is simple: the tokenized US CP tranche should leave the issuer with fewer recurring breaks, fewer manual interventions, and a cleaner control model than the existing process for the same funding objective.

7.2 Design what-if scenarios

Does the token layer improve the same US commercial paper funding workflow enough to reduce recurring reconciliations, manual confirmations, settlement exceptions, and lifecycle-processing friction beyond what the existing CP process or a non-token workflow upgrade would deliver?

- The pilot removes defined manual and reconciliation steps while preserving authoritative records and controls: Cost Savings and Risk Reduction become more durable.
- The token layer remains parallel to existing records and servicing: Tokenization Costs rise and the operational value case weakens.
- Cash-leg finality, failed-settlement recovery or responsibility allocation remains unresolved: New Risks should be treated as higher and scale-up should pause.

7.3 Suggested next steps

- Establish the conventional issuance and servicing baseline and pilot success thresholds.
- Document the authoritative record, token role, RACI, lifecycle event model and recordkeeping obligations.
- Validate DvP, cash-leg finality, failed-settlement, cancellation, rollback and recovery procedures end to end.
- Agree which legacy steps can be retired and tie provider economics and scale-up to measured evidence.

TVAC

Appendices

- Appendix A — Method & interpretation
- Appendix B — Operational roadmap
- Appendix C — TVAC Provider Directory (illustrative)
- Appendix D — Glossary & abbreviations
- Appendix E — Case inputs used for assessment
- Appendix F — Disclaimer



Appendix A — Method & interpretation

A.1 How to use this appendix

This appendix explains how to interpret the assessment, what evidence is still needed, and which conditions could strengthen or weaken the case. It is not a technical manual for TVAC.

A.2 What the assessment establishes — and what it does not

This report provides informational decision support only. It does not constitute legal, regulatory, tax, financial, or investment advice, and it does not provide a compliance determination, valuation, or provider endorsement.

Case snapshot (as stated)

- Issuance jurisdiction: United States
- Sale jurisdictions / target markets: United States
- Asset / instrument type (indicative): wrapper
- Liquidity intent (as stated): none
- Payment / cash-leg indication: T+1/T+2 securities settlement
- Cash-leg dependency / readiness: Critical dependency / Selected / Unverified readiness

What the Tokenization Value Verdict means

- Viable: Tokenization added value is plausible under the stated assumptions, subject to normal verification and execution.
- Conditional Go: Tokenization added value is design-dependent. Proceeding without meeting the listed conditions materially increases execution risk.
- Non-viable: Under current assumptions, tokenization is unlikely to outperform a conventional structure on a risk-adjusted basis.

What TVAC did not assess

- No definitive legal, regulatory, tax, or investment conclusions
- No asset valuation, yield projections, or market-price forecasts
- No assurance of secondary-market liquidity or investor demand
- No operational readiness audit beyond the indicative checklist

A.3 Minimum evidence before the next major commitment

Before proceeding toward implementation, decision-makers should be able to substantiate a minimum level of evidence across the areas below.

This checklist is not exhaustive and does not imply regulatory sufficiency. It reflects common governance expectations in institutional-grade projects.

Highest-priority gaps in this case

(derived from inputs/completeness)

1. No verified case-specific regulatory claim is activated; approved baseline context supports structured scoping, while case-specific legal application remains unresolved.
2. Obtain written legal and control analysis of the token layer, authoritative instrument and investor records, participant and intermediary roles, custody or registry responsibilities, settlement finality and the operating perimeter.
3. Document and validate the relationship between the tokenized workflow, authoritative issuer, agent, registry or custody records, security- and cash-leg finality, exception handling, rollback and recovery.

Structural & economic foundations

- Clear description of the underlying asset and ownership structure
- Defined economic rights and cash-flow logic
- Draft term sheet or structural outline
- Clear rationale for tokenization vs alternatives

Regulatory & distribution posture

- Issuance jurisdiction clearly defined
- Explicit list of sale jurisdictions
- Documented investor type assumptions
- Initial regulatory scoping (high-level)

Token & control design

- High-level token design and rights mapping
- Transfer restriction and whitelist logic (conceptual)
- Corporate actions and investor reporting approach
- Governance for smart-contract or protocol changes (if relevant)

Infrastructure & counterparties

- Custody model (conceptual)
- Indicative partner shortlist (non-binding)
- Issuance, registry, and lifecycle approach
- Payment / cash-leg and settlement approach, proportionate to the case's actual value proposition
- Early assessment of secondary trading (if any)

Cost & operational realism

- High-level CAPEX and OPEX view
- Identification of major cost drivers
- Clear internal ownership of ongoing operations
- Understanding of scalable vs non-scalable costs

Lack of clarity across several of these areas materially increases execution risk, even where the conceptual case appears attractive.

A.4 What could strengthen or weaken the case

Top drivers in this case (derived from scoring)

- Upside driver: Issuance and lifecycle improvement (value depends on measurable reductions in manual work, reconciliation, reporting and servicing)
- Risk driver: New risks (most sensitive to authoritative-record mapping, cash-leg finality, recovery design and provider-role clarity)
- Cost driver: Tokenization costs (most sensitive to fixed pilot costs, integration burden, duplicate processes and provider scope)

High-impact decision levers (practical)

- Legacy-process retirement: identify which allocation, reconciliation, reporting, coupon and maturity steps can be removed after validation.
- Authoritative record and controls: define the legally and operationally authoritative record, the role of the token layer and the responsibility split across the operating stack.
- Settlement and recovery: validate DvP sequencing, cash-leg finality, failed-settlement handling, cancellation, reversal, rollback and recovery.
- Provider economics: tie fees and scale-up to service levels, measurable process improvement, portability and incident performance.
- Reuse across later issuances: improve economics by reusing validated integration, governance and lifecycle capabilities rather than rebuilding them for each tranche.

How to use this

Improve the operating evidence and control model before adding distribution, fundraising, trading-support or liquidity objectives. Those objectives would materially change the case and require a separate assessment.

A.5 Evidence & source transparency

Basis of assessment

The assessment is based on the case description and structured inputs used for this assessment, the fixed TVAC methodology, and any activated verified external sources. Operational and commercial claims in the case inputs have not been independently verified unless explicitly stated.

External sources used

- SEC — Exempt Offerings overview · U.S. Securities and Exchange Commission · checked 2026-07-15
- SEC — Guide to Broker-Dealer Registration · U.S. Securities and Exchange Commission · checked 2026-07-15
- SEC — Alternative Trading Systems · U.S. Securities and Exchange Commission · checked 2026-07-15
- SEC — Transfer Agents · U.S. Securities and Exchange Commission · checked 2026-07-15
- DTCC — U.S. trades successfully processed using DTC-tokenized assets · Depository Trust & Clearing Corporation · checked 2026-09-14
- SEC staff — Statement on Tokenized Securities · U.S. Securities and Exchange Commission staff (Divisions of Corporation Finance, Investment Management, and Trading and Markets) · checked 2026-09-23

What remains unverified

- Which record is legally authoritative for the debt instrument and investor positions, and what legal and operational role does the token layer perform relative to that record?
- How are responsibilities divided among issuer, arranger, transfer or registry operator, custodian, wallet provider and settlement bank for issuance, position maintenance and lifecycle events?
- How are asset-leg and cash-leg finality, DvP sequencing, failed settlement, cancellation, reversal, rollback and recovery defined for the pilot?
- Which eligibility, onboarding, disclosure, communications, marketing and transfer-control requirements apply in each stated jurisdiction to the professional-investor perimeter?
- How must interest or discount, principal, maturity, corporate-action, tax and recordkeeping events be reflected across the authoritative record, token layer and existing issuer systems?
- Which legacy issuance and servicing steps may be retired after validation, and what evidence is required before parallel processing can be discontinued?

Appendix B — Operational roadmap

This roadmap translates the assessment into a staged implementation path for the specific case type. It is not a project plan or implementation instruction: each phase should be adapted to the final legal structure, counterparties, providers and evidence available. Progress only when the decision gate for the preceding phase has been met.

B.1 Operational roadmap (structured)

The items below are indicative planning scaffolding. They are not a substitute for jurisdiction-specific legal, tax, financial, and operational due diligence, and they must be tailored to the selected asset type, wrapper structure, investor segment, custody model, venue, and settlement design.

B.2 Phase 1 — Preparation & feasibility

Case-specific focus for this phase: Primary issuance jurisdiction: United States. Initial distribution scope: United States. Initial investor focus: professional investors, accredited / qualified investors.

Objectives

- Clarify the underlying asset/instrument, economic rights, and proposed token wrapper.
- Define target investor segments (professional / eligible / retail, if applicable) and distribution scope.
- Establish internal alignment on objectives, success criteria, and risk appetite.

Key decisions

- Token rights model (equity-like, debt-like, fund-like, revenue share, participation note, etc.).
- Transfer restriction regime (whitelisting, holding periods, jurisdictional gating).
- Custody and settlement direction (account-based vs. wallet-based; T+0 vs. T+1+).

Deliverables (checklist)

- One-page case definition: asset, wrapper, jurisdiction(s), investor segment, and target size.
- Preliminary term sheet outline (rights, restrictions, fees, lifecycle events, disclosures).
- High-level implementation estimate (time bands, cost bands, key dependencies).
- Initial risk register (top 10–15 risks with owners and first mitigations).

Primary stakeholders

- Issuer/asset owner (business sponsor, finance, legal/compliance, operations/IT).
- External legal counsel (securities, funds, structuring), tax adviser, and (if relevant) auditor.
- Shortlisted tokenization platform/technology provider, regulated custodian, and potential venue(s).

Dependencies / gates

- Clear legal characterization of the instrument and its regulatory perimeter in the issuance jurisdiction.
- Agreement on must-have constraints (investor eligibility, reporting, custody, settlement, data handling).
- Internal approval to proceed to structured design (budget and governance gate).

Risk hotspots

- Unclear instrument classification or incompatible distribution intent (e.g., retail scope without a verified pathway).
- Mismatch between user-stated transfer expectations and the legally and operationally supportable process.
- Hidden operational burdens (cap-table/admin, reporting, corporate actions, tax documentation).

B.3 Phase 2 — Legal, tax & regulatory structuring

Case-specific focus for this phase: Core legal and tax analysis should start from United States. Distribution permissions must be validated for United States. Proposed disclosure approach: Exempt. Applicable local requirements must be confirmed. Transfer-control design to validate: Whitelist + accredited/professional-only + issuer/agent-approved transfers.

Objectives

- Obtain independent legal validation of a legally supportable issuance and distribution structure for the intended jurisdiction(s).
- Define disclosure, governance, and ongoing compliance obligations for the lifecycle of the instrument.
- Obtain qualified tax analysis of the proposed wrapper and cash-flow mechanics for issuer and investors.

Key decisions

- Issuance vehicle choice (direct issuer vs. SPV; fund wrapper vs. corporate wrapper).
- Offering and documentation pathway verified for the stated jurisdiction(s); do not infer a legal category from interface labels.
- Investor onboarding model (KYC/AML responsibilities, reliance model, delegated vs. in-house).
- Choice of legal documentation stack verified by qualified local advisers.

Deliverables (checklist)

- Legal characterization memo and jurisdiction matrix (requirements by country/investor segment).
- Draft offering documentation and investor disclosures.
- Transfer restriction policy and rulebook (whitelist logic, jurisdictional gating, and any explicitly selected holding-period rules).
- Tax memo(s) covering withholding, reporting, and cross-border distribution considerations.
- Compliance operating model (roles, escalation paths, record-keeping, audit trail).

Primary stakeholders

- Issuer legal/compliance, external counsel, tax advisers.
- Custodian/compliance function (if regulated custody is used), transfer agent (if applicable).
- Any transfer provider or environment explicitly retained in scope and independently validated.

Dependencies / gates

- Independent legal confirmation of the offering pathway and distribution permissions across the stated target jurisdictions.
- Operational feasibility of investor onboarding and transfer controls (including data privacy constraints).
- Formal sign-off on documentation pack and compliance model.

Risk hotspots

- Jurisdiction conflicts (marketing rules, suitability requirements, local registration triggers).
- Under-scoped ongoing obligations (periodic reporting, investor communications, corporate actions).
- Tax frictions that erode the added value thesis (withholding, reporting, complexity costs).

B.4 Phase 3 — Build, integration & vendor onboarding

Case-specific focus for this phase: The current custody design indicates regulated custodian. Its precise legal and operational role remains to be confirmed. The current transfer environment indicates controlled OTC transfers; this does not establish a local legal venue classification. The current settlement design indicates T+1/T+2 securities settlement. Its operational and legal fit remains to be validated. Transfer-rule implementation must reflect: Whitelist + accredited/professional-only + issuer/agent-approved transfers.

Objectives

- Finalize the technology stack and integrate custody, settlement, onboarding, and reporting workflows.
- Implement transfer rules and lifecycle event handling (issuance, redemptions, distributions, corporate actions).
- Establish operational readiness (controls, monitoring, incident response, reconciliation).

Key decisions

- Token standard / ledger choice and compatibility with custody and venue requirements.
- Settlement model (atomic delivery-versus-payment vs. coordinated settlement through intermediaries).
- Data and reporting architecture (who stores what; audit trail; privacy boundaries).
- Outsourcing vs. in-house responsibility split (transfer agent, admin, reporting).

Deliverables (checklist)

- Final technical architecture and integration plan (custody, venue, payments, identity/KYC).
- Rule engine / whitelist implementation and test suite (including jurisdictional gates).
- Operational runbooks (reconciliation, incident response, corporate actions, investor support).
- UAT results and sign-offs (issuer, custodian, venue, admin/TA as relevant).
- Security review (key management model, access control, change management, monitoring).

Primary stakeholders

- Issuer operations/IT, tokenization platform/tech provider, regulated custodian where explicitly selected.
- Selected transfer-environment integration team, payment rails (if applicable), admin/TA providers.
- Compliance and risk functions for control design and approval.

Dependencies / gates

- Signed vendor contracts, SLAs, and responsibility matrix (RACI) across the stack.
- Completed integration testing and reconciliation proof (end-to-end lifecycle flows).
- Readiness gate: controls, monitoring, and support operations in place.

Risk hotspots

- Integration complexity and reconciliation gaps across custody/venue/admin systems.
- Rule enforcement drift (transfer restrictions not consistently applied across venues/wallets).

- Operational single points of failure (provider concentration, key management, support load).

B.5 Phase 4 — Issuance, distribution & launch

Case-specific focus for this phase: Any legally required launch permissions and onboarding constraints must be independently confirmed for: United States. Intended investor group for launch planning: professional investors, accredited / qualified investors. Eligibility under local law must be confirmed. The launch transfer environment currently indicates controlled OTC transfers; this does not establish a local legal venue classification. The current liquidity intent is no secondary liquidity planned. No market status, matching demand or liquidity support is inferred.

Objectives

- Execute issuance mechanics and investor onboarding under the independently verified offering pathway.
- Operationalize disclosures, subscriptions, allocations, and post-issuance record-keeping.
- Operationalize only the user-stated and legally validated transfer process, if any; no venue listing, market making or liquidity support is assumed.

Key decisions

- Distribution channel mix (direct, intermediated, platform marketplace, placement agents) as legally validated.
- Subscription mechanics and settlement timing (funding rails, DVP, cutoff times) as actually specified.
- Transfer model, if any: stated approvals, eligibility, timing and limitations; no cadence or support mechanism is assumed.

Deliverables (checklist)

- Final investor communication pack and distribution checklist.
- Subscription and allocation runbook; reconciliation and exception handling.
- Go-live monitoring plan and incident escalation paths.
- Documented transfer-process setup, if applicable, and evidence-based monitoring metrics.

Primary stakeholders

- Issuer distribution team, platform operator, onboarding/KYC provider.
- Custodian, settlement and transfer providers explicitly selected for the stated process.
- Legal/compliance for marketing controls and ongoing disclosure obligations.

Dependencies / gates

- Approved marketing/distribution materials and compliance sign-off.
- End-to-end dry run of subscription, allocation, and post-issuance reporting.
- Operational readiness gate: support, monitoring, and governance in place.

Risk hotspots

- Distribution frictions (slow onboarding, suitability issues, cross-border marketing constraints).
- Unsupported transfer or liquidity expectations relative to the actual process and evidence.
- Operational load spikes (investor support, reconciliation exceptions, reporting deadlines).

B.6 Phase 5 — Post-issuance operations & monitoring

Case-specific focus for this phase: Ongoing governance and evidence trails must remain defensible under United States. Post-issuance compliance must continue to support holders across: United States. Ongoing operations must reflect the regulated custodian design once its precise role has been independently confirmed. Post-launch monitoring must reflect the actual validated transfer process currently described as controlled OTC transfers.

Objectives

- Run ongoing compliance, reporting, and investor communications through the instrument’s lifecycle.
- Operate lifecycle events (distributions, redemptions, corporate actions) with strong controls and auditability.
- Monitor provider performance, costs, and risk posture; adjust as regulations and market conditions evolve.

Key decisions

- Governance cadence (board/committee oversight, change approvals, vendor reviews).

- Reporting scope and frequency (investor reporting, regulatory reporting, audit readiness).
- Contingency planning (provider exits, custody changes, venue strategy shifts).

Deliverables (checklist)

- Ongoing reporting calendar and control framework (incl. audits and reconciliations).
- Lifecycle event runbooks and evidence retention procedures.
- Vendor performance KPIs and incident log; periodic risk reviews.
- Contingency and change-management playbook (custody/venue migrations, rule updates).

Primary stakeholders

- Issuer operations, compliance, finance; custodian; admin/TA (where applicable).
- Transfer and settlement providers explicitly retained in scope, if any.
- External auditors/advisers for periodic reviews and regulatory updates.

Dependencies / gates

- Stable operating model and evidence trail sufficient for audit/regulatory scrutiny.
- Clear change-management process for token rules, disclosures, and provider configuration.
- Budgeting and resourcing for ongoing operations (often underestimated in early planning).

Risk hotspots

- Compliance drift as rules change or distribution expands across jurisdictions.
- Operational fragility under low-liquidity or stress scenarios (settlement failures, disputes).
- Provider concentration risk and cost creep relative to value delivered.

Appendix C — TVAC Provider Directory (illustrative)

This appendix is an illustrative reference layer intended to help the reader think about which types of external partners may be relevant in a tokenization project.

It is not a recommendation list, procurement shortlist, or ranking. Inclusion or omission does not imply endorsement, criticism, quality judgment, regulatory approval, or case suitability. Any provider mentioned here must be assessed independently for legal fit, jurisdictional coverage, operating model fit, technical capability, conflicts, cost, and execution reliability. This appendix is non-exhaustive and subject to change.

C1. How to use this appendix

Use this appendix to identify the provider categories a case may require, not as a signal that a specific provider should be engaged.

In many cases, the main value of the directory is not the names themselves, but the discipline of asking:

- Which functions must be covered?
- Which functions can be combined?
- Which functions must remain regulated or jurisdiction-specific?
- Which dependencies create concentration risk or vendor-capture risk?
- Which parts of the stack are critical before launch, and which can wait?

C2. Typical provider categories

Depending on the case, a tokenization project may require some or all of the following partner categories:

- legal & regulatory advisers
- tax advisers
- issuance / tokenization platform
- smart-contract / token engineering support
- custody / wallet infrastructure
- transfer agent / registry administration
- fund administration or SPV administration
- trading venue / ATS / MTF / exchange partner
- transfer or liquidity-support provider, only where explicitly required and legally permitted
- KYC / AML / onboarding provider
- payment / settlement / banking partner
- audit / assurance / cybersecurity reviewers
- valuation, data, or oracle-related providers
- distribution / placement / investor onboarding partners
- systems integration / workflow automation support

C3.1 Legal & regulatory advisers

These advisers help determine whether the proposed structure is legally enforceable, properly classified, and consistent with the intended issuance, distribution, transfer, custody, and disclosure model.

Typical use cases:

- wrapper design
- securities-law classification
- cross-border marketing analysis
- transfer restriction design
- offering documentation
- legal opinions and regulatory scoping

C3.2 Issuing platforms

These providers typically support issuance workflows, lifecycle management, permissions, investor access rules, and parts of the reporting or compliance stack.

Typical use cases:

- token issuance
- lifecycle events
- permissions / whitelisting
- cap-table or investor-position management
- compliance rule integration
- reporting workflows

C3.3 Custody, trading & settlement

These providers support safekeeping, wallet infrastructure, execution, venue access, settlement, and post-trade operations.

Typical use cases:

- institutional custody
- secure wallet infrastructure
- settlement support
- exchange / venue access
- post-trade servicing
- collateral mobility or transfer control support

C4. Practical selection principles

When evaluating providers, the reader should typically assess:

- jurisdictional fit
- regulatory status where relevant
- experience with the relevant asset class
- operating-model compatibility
- integration burden
- exit options / portability
- fee model
- service resilience
- auditability and evidence quality
- concentration risk

Where possible, provider choices should reinforce the case logic rather than undermine it. A weak provider stack can easily erode the value thesis of an otherwise promising structure.

C5. Important caution

The right provider stack is highly case-dependent.

A provider suitable for one asset class or operating model may not fit another. A provider strong in digital-asset custody may not be the right choice for transfer administration or legal structuring. A venue that looks attractive commercially may create regulatory or operating-model problems if the case does not actually need that level of transferability.

Appendix D — Glossary & abbreviations

A practical reference for the most common terms and abbreviations used in this report. Spelling note: tokenization uses “z”.

Term	Meaning
Accredited investor	An investor category used primarily in the United States for persons or entities meeting specific income, net worth, or institutional eligibility thresholds for participation in certain exempt offerings.
Added Value (AV)	TVAC’s net assessment of whether tokenization appears to improve a case overall once both upside and friction are weighed together. Formula: AV = (NO + CS + RR) – (TC + NR).
Asset owner / Issuer	The originating party, sponsor, manager, or issuing structure responsible for launching or sponsoring the tokenized case being assessed.
ATS	Alternative Trading System. A regulated trading venue category used in some jurisdictions, particularly the United States.
Cash leg	The cash side of a transaction or settlement flow, as distinct from the asset or token side.
Conditional Go	A Tokenization Value Verdict indicating that tokenization may add net value, but remains materially dependent on design discipline, implementation choices, and resolution of key gating conditions.
Custody	The safeguarding, control, and recordkeeping of assets or positions on behalf of investors or market participants.
Distribution scope	The intended investor reach of an offering or tokenized structure, including geography, investor type, and access conditions.
DvP	Delivery versus Payment. A settlement model in which transfer of the asset and transfer of payment are linked so that one does not occur without the other.
Economic rights	The financial entitlements attached to a token or instrument, such as income participation, coupon, principal repayment, or profit-sharing.
Execution risk	The risk that a concept that appears viable in theory fails in practice because implementation, coordination, economics, or governance proves weaker than assumed.
Governance	The decision-making, control, accountability, and escalation framework surrounding a structure, platform, issuer, or operating model.
Investor segmentation	The way the target investor base is defined and limited by investor type, sophistication, jurisdiction, or eligibility criteria.
Net Added Value	See Added Value (AV).
New Opportunities (NO)	Case-specific upside that tokenization creates or materially strengthens, such as new investor reach, new operational utility, collateral use, programmability, or improved strategic flexibility.
New Risks (NR)	Additional risks introduced by tokenization, including legal, regulatory, operational, technological, governance, distribution, or market-structure risks.
NO / CS / RR / TC / NR	Shorthand for TVAC’s five factors: New Opportunities, Cost Savings, Risk Reduction, Tokenization Costs, and New Risks.
Non-viable	A Tokenization Value Verdict indicating that the current structure does not support a positive tokenization value case under the TVAC methodology and should not proceed without substantial redesign.
Parallel processes	A situation in which tokenized and legacy workflows both remain active, limiting the real efficiency gains from tokenization.
Provider economics / value-capture misalignment	The risk that provider fees, integration burden, contractual dependencies or operating responsibilities absorb or displace the measurable value that the initiating institution is expected to capture from the tokenized workflow.
Registry	The official or operational record of holdings, ownership, or investor positions.
Retail investor	A non-professional investor who generally receives higher levels of regulatory protection than professional or institutional investors.
Rights mapping	The process of making clear what legal, economic, governance, and transfer rights a token actually represents and how those rights are enforceable.
Risk Reduction (RR)	The extent to which tokenization reduces existing risk in the conventional setup, for example through better controls, improved transparency, clearer auditability, or lower reconciliation burden.
SEC	Securities and Exchange Commission in the United States.

Term	Meaning
Self-custody	A model in which the end holder directly controls the relevant wallet, keys, or token position without relying on a traditional third-party custodian.
Settlement model	The way asset transfer, cash transfer, record updates, and finality are operationally achieved.
SPV	Special Purpose Vehicle. A ring-fenced legal entity often used for issuance, structuring, asset isolation, or liability segregation.
Tokenization Costs (TC)	The additional costs introduced by tokenization, including structuring, legal work, compliance, integration, platform costs, governance overhead, servicing changes, and operating complexity.
Transfer controls	Rules, permissions, and technical or legal mechanisms governing who may transfer, receive, or hold the token.
Transfer restrictions	Specific limitations placed on secondary transfers, often for regulatory, contractual, or investor-eligibility reasons.
TVAC	Tokenization Value-Add Calculator. A structured decision-support methodology for assessing whether tokenization appears to create net added value in a specific case.
Vendor-capture risk	The risk that a structure becomes economically or operationally overly dependent on one platform, provider, or implementation path in a way that shifts value away from the initiating party.
Viable	A Tokenization Value Verdict indicating that the TVAC methodology identifies a positive tokenization value case under the described structure, subject to validation and disciplined execution. It is not an investment, credit, yield, valuation, legal, tax, or suitability assessment.
Whitelist / Whitelisting	A mechanism for restricting participation or transferability to pre-approved investors, wallets, counterparties, or participants.

Appendix E — Case inputs used for assessment

E.1 Full verbatim case description

Full original free-text case input provided by the user.

We are a US-based corporate issuer with an existing, recurring commercial paper (CP) program used for short-term working capital. Today the program relies on multiple intermediaries, manual confirmations, repeated reconciliations between the issuer's treasury team, issuing/placing agents, custodians, and investors, and a settlement cycle that creates operational friction and occasional settlement fails.

We want to introduce a tokenized CP note format (economically equivalent to our existing CP) for professional and accredited investors only. This is not a retail product and not a public exchange initiative. The scope is deliberately conservative: a single US issuance + distribution perimeter, whitelisted participation, and institutional-grade custody. The primary objective is post-trade efficiency and risk reduction rather than "new liquidity".

The tokenized format is designed to (1) reduce reconciliation breaks by moving to a single shared position record, (2) automate issuance and lifecycle events (issuance confirmations, coupon/accrual reporting, maturity repayment notices), and (3) tighten settlement controls through delivery-versus-payment (DvP) style workflows. Importantly, we commit to decommissioning duplicated legacy steps for the tokenized tranche (no permanent parallel "double-booking"), so savings are structural rather than cosmetic.

We expect measurable cost savings (reduced agent/custody operations per issuance, fewer manual exceptions) and measurable risk reduction (fewer settlement fails, cleaner audit trail, reduced counterparty/operational exposure). Any secondary transfers, if permitted at all, are tightly controlled and limited to eligible whitelisted holders under standard securities restrictions. The issuer is the primary beneficiary of the operational savings and control improvements under this design.

E.2 Structured input record

This record reproduces the structured inputs supplied for this assessment. For dropdowns and checkboxes, it also shows the alternatives that were available but not selected.

Field	User response	Other available options not selected
Issuance jurisdiction	United States	Not applicable
Sale jurisdictions (comma-separated)	United States	Not applicable
Instrument type	Digital wrapper (CSD/TA/registrar)	Sole-record / native tokenized security
Custody model	Regulated custodian	Transfer agent / CSD; Self-custody (wallets)
Secondary-market liquidity	None	Ad-hoc OTC; Quarterly windows; Continuous
Secondary-market venue	OTC	ATS / MTF; DEX (permissioned); CSD / TA-registered
Payment / cash-leg model	T+1/T+2 securities settlement	Not decided yet; Not material at this stage; Traditional bank / fiat payments; Custodian / bank / TA / CSD settlement; Stablecoin / tokenized cash; Deposit token / tokenized deposit / institutional DvP rail; Advanced: T+0 / atomic DvP; Advanced: off-chain booking
Offering disclosure	Exempt / limited-offering route (jurisdiction-dependent)	KID (PRIIPs); Prospectus
Transfer restrictions	Whitelist + accredited/professional-only + issuer/agent-approved transfers	Not applicable
Tax considerations (optional)	Standard US withholding and investor reporting depend on holder type; tax handled under existing CP program processes.	Not applicable
Investor types	Professional; Qualified / accredited (jurisdiction-dependent)	Retail

An option shown as not selected was not necessarily actively considered or rejected by the user; it was simply not selected for this assessment.

> **Scenario comparison:** Alternative options shown here can help identify design variables worth testing in a new assessment or in TVAC Scenario Lab—for example a different custody model, settlement model, participant type or transfer setup.

Appendix F — Disclaimer

Disclaimer

This report is provided for informational and decision-support purposes only. It does not constitute legal, regulatory, tax, accounting, investment, valuation, or financial advice, and it should not be relied upon as a substitute for professional advice tailored to the specific facts and circumstances of the case.

TVAC is a structured assessment tool designed to evaluate whether and when tokenization may create added value relative to conventional alternatives. Its outputs reflect a methodology-based analytical framework operating under uncertainty. They are not guarantees of commercial success, feasibility, investor demand, regulatory approval, liquidity, execution quality, budget outcome, or risk reduction.

Any scores, verdicts, scenarios, stakeholder assessments, suggested actions, implementation observations, and roadmap elements in this report are indicative only and depend on the quality, completeness, and interpretation of the input data, as well as on assumptions about structure, jurisdiction, service-provider readiness, market conditions, execution capability, and timing.

Tokenization outcomes can differ materially from this assessment if key conditions change, including legal structuring, investor scope, jurisdiction count, custody model, distribution pathway, settlement design, governance arrangements, servicing readiness, technology integration, cost allocation, evidence quality, or liquidity support.

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TVAC

Tokenization Value-Add Calculator

A PRODUCT BY ASTOVAL

Astoval provides independent, AI-enhanced early-stage decision support for asset tokenization, helping organizations assess whether tokenization creates genuine added value before significant resources are committed to legal structuring, platform selection or implementation.

PRIVACY AND DATA

Case inputs and generated reports are processed for delivery and are not intentionally retained as a persistent customer-case record in the standard self-service flow. OpenAI API data is not used for model training unless the API customer opts in; separate abuse-monitoring retention may apply for up to 30 days.

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